DECO2300/7230 Digital Prototyping and Extended Reality: Prototype 1

Brief Description of the assignment

You will create and test a low-fi rapid prototype of the app you have chosen to support in an extended reality environment. This process will involve ideation (sketches, storyboards), low-fi rapid prototyping (mock-ups with paper, cardboards, playdoh etc.) to play-act/enact the interactions, and testing to get feedback on your concept. Remember: you are working on a concept, on some immersive interaction ideas to support some of the users' tasks and goals—you are not working towards a final product with all the functionalities. Your prototype is evaluated with your peers in week 4 and the report is due the following Monday on week 5. The tasks involved in this assessment are detailed below.

Prototype 1 – Detailed tasks & requirements

1- Choose an application and identify three tasks/interactions with this app. Describe which immersive, extended reality interactions and affordances will be used to support these tasks in an XR environment. **Note: the tasks and interactions should not be basic, e.g., limited to pushing VR buttons.**
2- Define a concept: your vision of how this app is translated to an immersive environment. What is involved in terms of environment, spatial interaction? Which specific interactions and features you want to test to validate part of your concept? Sketches and storyboards are expected for this.
Reminder: you are not building the whole new app; you are focusing on key design and interactions of the concept.
3- Create a low-fidelity rapid prototype with paper, carboard, foam, playdoh etc. This low-fidelity, rapid prototype should allow you to explore and get feedback on key tasks/interactions involved in your concept. It is not required that this prototype works from a functional perspective. It allows you to test ideas, concepts and tasks, rapidly.
4- Test your prototype in week 4 during studio with your peers and collect feedback.
5- Report on your testing session and design process.

Delivery stages and dates:

- **Week 4 (just after your allocated studio session):** upload photos of your rapid prototyping material (storyboard, carboard, foam etc.) at the end of the testing session in your studio (Uploads up to 200MB). Talk with your demonstrator if you have special needs.
    - **ATTENDING TESTING SESSIONS ARE COMPULSORY**
    - **YOUR PARTCITIPATION IN THE USER TESTING SESSIONS WILL ALSO BE MARKED.**
    - **NOT ATTENDING WEEK 4 STUDIO WILL RESULT IN A FAIL FOR PROTOTYPE I.**
    - **ATTEND YOUR ALLOCATED STUDIO FOR THE TESTING SESSION**
- **Week 5**: statement of delivery submission (see below). Due date: **19/08/2024 (Monday week 5) at 15:00**

Statement of Delivery requirements (Monday Week 5)

The document needs to include the following content:

• **Basic Details:** Full name, student number, practical session & Demonstrator name & document title.

• **Statement of Originality:** a statement identifying and acknowledging the sources of content used in your prototype (see the accompanying document Statement of Originality for how to do this).

• **Your concept** – describe your concept for the application and the three tasks you chose. How is the user interacting in XR with this application? Sketches, storyboards and explanatory text are expected to explain your concept. This should demonstrate the intended key interactions with your XR concept.

• **Testing Agenda/Protocol** – You should address:

- what is the purpose of the testing session?
- What questions, issues, or information about specific design considerations you are seeking?
- What aspects of the concepts you intend to test using your rapid prototype?
- What feedback you are after?

• **Evaluation Method** - how you are going to conduct the evaluation so that you collect the information you need. Your Statement of Delivery will document this information, describing the low-fi prototype, how it allows you to achieve your testing session and your evaluation outcomes.

• **The Purpose of this testing round:** A description of the aspect/s of the concept that are to be evaluated and what you hope to find out through people using the prototype. Including a discussion of what was important to include & exclude in the construction of the prototype, to meet the requirements of the evaluation.

• **The Form of the prototype:** A brief description of the prototype itself. Remembering that low-fi, rapid prototypes are not representative of the finished product. Screenshots, photographs of physical artefacts, description of flow of interaction should be included here - these should also be accompanied by explanatory text and/or annotations.

• **Evaluation Outcomes & Reflection:** What were the results of the evaluation? What conclusions can you draw about the concept from the results? The feedback from your participants after the testing session should be summarised & presented as an overall analysis. If applicable, describe the changes you will make to your concept in response to the evaluation. And finally, discuss whether the prototype you created was sufficient to test what you wanted to test.

• **Testing Plan:** Based on what you have learned in this first testing session, briefly describe what aspects of your concept (e.g., which XR interaction, design worked best) you intend to test with the follow up hi-fidelity Prototypes 2 and 3.

• **Miro Link** (or other electronic journal): Please provide a link to your miro board allowing edit privileges

• **Video Link to your prototype**: Submit a link to a short video (3-5 minutes) that demonstrates your low fidelity/rapid XR prototype (see slides 55 to 59 in Lecture 2 for reference). Your video should clearly show your key concept and immersive interactions and illustrate how users interact with your XR design. Remember that you can use short edits, enacting and Wizard of Oz to simulate XR elements and interactions. The video doesn't have to look professional, but it should effectively communicate your ideas and demonstrate the XR affordances you've incorporated in your design. Prioritise clarity and demonstration over polish.

Marking Criteria

You will be graded according to the following criteria:

- Concept and XR Interaction Design
  - Concept is clearly articulated. Key XR interactions are well-defined and align strongly with user tasks. Immersive design elements and affordances are thoughtfully incorporated.
- Design Process and Prototyping
  - Relevant and detailed sketches and storyboards. Low-fi prototype effectively demonstrates key interactions. Low-fi prototype materials used to simulate XR environment and interactions in a meaningful way.
- Testing and Evaluation
  - Clear testing agenda with specific, relevant questions. Evaluation method well-suited to prototype and testing goals. Thoughtful reflection on testing outcomes.
- Analysis and Iteration Planning
  - Insightful, well summarised analysis of test results. Clear, justified plans for changes based on feedback. A thought-out plan is well described for future prototypes.
- Documentation and Presentation
  - Comprehensive, well-organised documentation. Video clearly demonstrates prototype and interactions. Miro board (or equivalent) is well documented and shows the design journey.

Submission information

- Submit a PDF file for your report
- Do not exceed 2000 words for your report (excluding the statement of originality, title, student name and details; add any relevant information into an appendix if needed). Be concise in your descriptions, bullet points are fine.